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PRO Research Process 4Ing: Project- and RDM-Oriented Research Process for Engineering Sciences

**Validation of a Newly Proposed Process to Combine Data Life Cycles and
Project-Oriented Research Processes in Engineering Sciences**

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Abstract

In a first paper Hamann, Robrecht, et al. 2024, we have shown the absence as well as the importance of a project-oriented [research data management \(RDM\)](#) process for engineering sciences. We argue that the integration of [RDM](#) into the practice would greatly benefit by the availability of such project-oriented [RDM](#) process. While many [RDM](#) and engineering research processes already exist, a combination of both was only recently designed in our previous paper Hamann, Robrecht, et al. 2024. Seeking for validity of the proposed RDM process, the research question in this article is: “Can our proposed [RDM](#) process be deployed into engineering research projects?” We interviewed ten (n=10) experts on their research practices and the fit of the proposed process to their research activities. Overall, we found a broad acceptance of the proposed process, as well as several possible potentials for enhancements, e.g. regarding wording and loops. As a result, this paper presents a validated project-oriented [RDM](#) process for engineering sciences.

1 Introduction

Integrating [RDM](#) processes into day-to-day work is still a hurdle for researchers in engineering sciences. Wuchner et al. 2023; Orloff et al. 2023 We showed the reasons for that in our previously published paper *Matching Data Life Cycle and Research Processes in Engineering Sciences* Hamann, Robrecht, et al. 2024: While many suggestions on how to carry out [RDM](#) already exist ELIXIR-CONVERGE 2023; Politze 2019; Higgins 2008, their integration into practice is not feasible on a one-to-one basis focusing on a strong theoretical reference Cox and Tam 2018. In this context, the data life cycle (DLC) model is often used to refer to only [research data](#). However, it cannot be transferred to research activities in general, yet it is frequently used for this purpose Hamann, Robrecht, et al. 2024; Cox and Tam 2018. Furthermore, for engineering sciences, [project](#)-orientation and specificity in engineering are missing in [RDM](#) processes. [Project](#)-oriented research has been increasing rapidly for years Krull and Tepperwien 2016; Winterhager 2015; Bonn 2009; Berning and Falk 2005, which is enforced by the “greater third-party funding activity in the natural sciences and engineering” Berning and Falk 2005. This [project](#)-oriented approach spans through all forms of research and occurs in basic research as well as in applied research. Behlau 2017

In our first paper Hamann, Robrecht, et al. 2024, we addressed this challenge (lack of integration of [RDM](#) into day-to-day work and [project](#) structures) while also focussing on the demands of researchers in engineering sciences. By that we identified two main issues with the state of the art in [project](#)-oriented [RDM](#) processes. First, while the variations of detail and complexity in the processes as well as the definition of their sub-steps and activities are vast, there is not yet a process with a scope, that reaches from the [project](#) as a whole down to the data in detail.

Second, there is a gap between DLCs and other, more holistic models. Hamann, Robrecht, et al. 2024 While DLCs support the management of data along its life cycle, they are not supposed to be utilised as a support structure for [project](#) management. DLCs are a “useful metaphor, but tend to encourage thinking that research processes are highly purposive, uni-directional, serial and occurring in a closed system. Research is often not like this” Cox and Tam 2018. Contrasting this insufficiency, engineering research processes often lack [RDM](#) context and applicability. Hence, a process combining both [project](#)-oriented research and [RDM](#) did not exist to the best of our knowledge. We aimed to close this gap with our proposed process.

The [finding](#) of this first paper Hamann, Robrecht, et al. 2024 is a multi-layered [RDM](#) process, combining the overall structure of [project](#)-oriented research with the [execution](#) of research and the handling of data as depicted in figure 1.

On the highest level, the [project management level](#), waterfall-like approach divides the research project into three phases. In the second phase, the [execution](#), the actual research takes place in form of the [content-wise execution](#). Within this [content-wise execution](#) the lowest level of research is found, where data collection is planned, conducted, checked for sanity and the data is stored for further use. This lowest level of research resembles the concept of DLCs and creates an [artefact](#). One or more artefacts

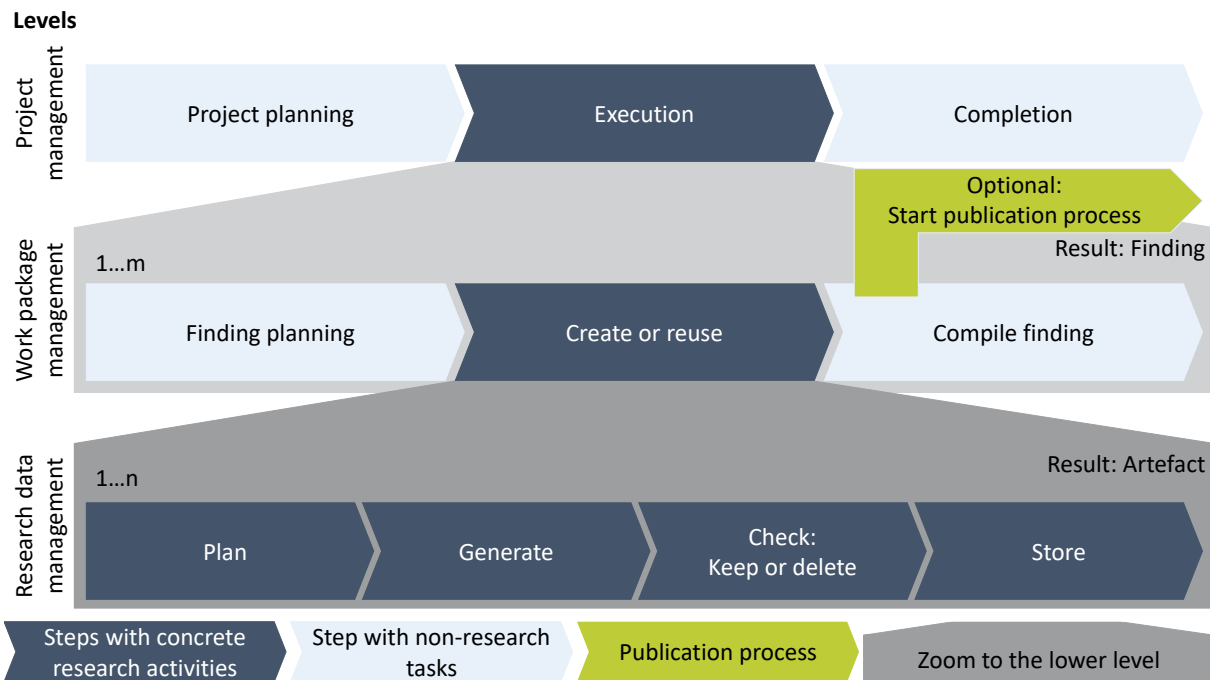


Figure 1: General structure of the previously designed RDM process Hamann, Robrecht, et al. 2024

can be compiled to a [finding](#) to answer a research question or complete a work package of a research project. One or more findings will lead to the goal of the project. For further detail on the process, please refer to appendix [A.2](#) or our first paper Hamann, Robrecht, et al. 2024.

While this process was built upon two data collections of researchers' practices and demands, the resulting process has to be cross-checked by its users to ensure that it is sufficient in filling the identified gap. On this basis, the applicability of the process has to be evaluated and validated in this article, leading to the research question:

Can our proposed RDM process be deployed into engineering research projects?

To answer the proposed research question, we conducted ten qualitative expert interviews. Experts were questioned on their day-to-day research activities and the fit of the proposed process – later called Project- and RDM-Oriented Research Process for Engineering Sciences (PRO Research Process 4Ing) – to these activities. It has to be mentioned that this article only describes the validation of the process itself as an “artificial evaluation” Venable 2006 to prove its fit to the proposed field of engineering sciences. The validation of the framework that inherits this process in a real world application was conducted by Hamann et al. Hamann, Florides, et al. 2025 in a Proof-of-Concept-Study.

The further structure of this paper is divided into four sections. Firstly, in section 2 we elucidate the [research methodology](#) used to validate and enhance our process. Secondly, the results of the validation are shown in section 3, which describes the collected feedback to the [previously designed process](#) for the possible enhancements. After that, section 4 discusses the methodology and limitations of the validation. Lastly, the paper closes with a [conclusion and discussion](#) (section 5).

2 Research methodology

To answer the proposed research question, we conducted expert interviews and analysed using qualitative content analysis. We conducted ten interviews, transcribed, anonymised, coded and aggregated. In the following, the exact procedure is explained.

2.1 Collecting the data: Planning and conducting the interviews

Expert interviews were conducted to validate the process. Interviews in general are a qualitative method of data collection that originates from the social sciences. Bogner et al. 2009; Froschauer and Lueger 2009 The question posed was processed using non-standardised interviews - more precisely, guided interviews. In this approach, interviewers use a guideline that comprises topics and questions they are expected to introduce during the interview. In order to create a natural flow of conversation, they can use the guidelines flexibly, adapt the order of questions and ask follow-up questions. This should allow the interviewees to answer as freely as possible. The expert interview method was chosen because a special type of knowledge was required for the validation: not only specialised knowledge, but also practical knowledge. In this type of interview, the experts themselves are not the object of investigation, but the medium through which the information for answering a question is collected. Gläser and Laudel 2010 The guideline for the expert interviews has been designed following the remarks of Gläser and Laudel Gläser and Laudel 2010.

Our goal was to have at least one interviewee - resp. expert - per TU9 – German Universities of Technology e.V. (TU9)-university, the Alliance of leading Universities of Technology in Germany, representative of engineering research at universities in Germany. In order to find experts who actually have the type of knowledge required, there were requirements for the interviewees: As the proposed process focusses mainly on engineering sciences, researchers had to have a corresponding engineering background. Furthermore, we demanded at least four years of experience in research. After some enquire, we could reach our goal of one interviewee per TU9-university with the required background.

The created interview guideline contained five parts (see table 1).

Part of the guideline	Content of the part
Part 1: Experiences with RDM	Previous experiences and attitudes towards RDM to contextualise the answers
Part 2: Presentation of the process	Interviewees were allowed to ask questions about the process and clarify ambiguities before the discussion
Part 3: Survey on the fit of the process to the interviewees' own research	Interviewees outlined the example project and were asked about the fit in this project for each process level as well as the publication process and the overall process. They first freely described how the research was conducted, after which the process was shown and they were asked for concrete comparisons to their descriptions.
Part 4: Information on the interviewees	A few demographic questions about the interviewees' careers and specialised contexts
Part 5: Conclusion	Space for concluding remarks and questions as well as an outlook on the further validation process

Table 1: Structure of the interview guideline

The free descriptions of their research activities were intended to ensure that the interviewees did not orientate themselves to the process before the interviewers asked for direct comparisons. In this way, the actual state was queried and not implicitly associated with a target state. Nevertheless, it was necessary to make the process accessible to them before the interview so that they could prepare themselves. The demographic section (part 4) asks respondents to categorise their technical and professional situation so that their answers can be put into context.

This guideline was reviewed by an expert for survey methods from the Chair of Intelligence in Quality Sensing (WZL-IQS) at the RWTH Aachen University (RWTH) and adapted accordingly. Additionally, a pretest was performed with a researcher from the RWTH. The learnings of the pretest were considered when conducting further interviews: minor improvements in the technical setup, interviewer behaviour through gaining experience, minor changes in the wording of questions. However, as these learnings were not focussed at the core content of the guideline, the pretest was still considered usable and was also included into the final results.

The interviewees received for preparation an introduction to the interview and the declaration of consent to the interview as well as a description of our proposed RDM process in form of an excerpt from the previous article about it Hamann, Robrecht, et al. 2024. They were asked to sign the declaration of consent, read the article and to think about a research project in which they were involved, ideally from the application phase to the project completion. If they were not involved in a research project from start to finish, they were supposed to choose a research project in which they were involved for as long as possible and in which they experienced either the application phase or the project completion.

The interviews lasted approximately one and a half hours each and were conducted by two interviewers simultaneously, one researcher and one member of the research supporting staff. This made it possible to divide the interviews according to background knowledge and to ask differentiated questions. The interviews were conducted online and recorded.

2.2 Preparing the data: anonymization and stock of data

After each interview, the recording was transcribed with the support of a custom python pipeline for audio transcription. The transcription was anonymised partly automated using QualiAnon¹. Mozygamba and Hollstein 2023 Afterwards, the anonymised transcripts were imported into MaxQDA².

Ten interviews were included in the subsequent analysis: one from each of the TU9 Universities and the pretest at RWTH Aachen University. The interviews contain a total of 117,066 words.

The demographic questions make it possible to categorise the interviewees in terms of their technical and professional backgrounds: The interviewees' scientific carriers are mostly set in mechanical engineering, but also in industrial engineering and economics. Data science and civil engineering are also represented. This means that the engineering sciences are thematically diverse in the interviews, which increases the quality of the validation. The interviewees already have a doctorate degree or are about to complete it. They have experience as research assistants, but also in research and transfer management. Some also fulfil management roles such as project and group leaders. Also, one professor was participating in the interviews. The minimum required four years of professional experience was mostly exceeded, in some cases significantly with eight years or more. These professional backgrounds guarantee the availability of the required expert knowledge and enable differentiated perspectives on the process.

All interviewees had already had contact with RDM. Their activities were partly of an administrative nature, partly they had used it themselves - both in solo work while writing their dissertations and in projects of different sizes. In the discussions, they mentioned many aspects of RDM that they had already encountered: archiving (hardware and software), publication, creating guidelines, Documentation, data management plan (DMP)s. They often did not work systematically, but intuitively and according to the "trial and error" principle, because knowledge was often lacking. Due to this lack of knowledge, tools that are not yet fully developed, external and internal guidelines that must be adhered to and poor integration into day-to-day work, their experiences with RDM have so far been predominantly bad. However, there are also interviewees who have had good experiences because they have benefited from the RDM activities after enduring initial stress.

Also, the interviewees were asked to bring an example project which would function as a mental guideline to describe their research processes. These projects were also surveyed in the interviews. These ranged from very personal projects with only a single person involved, e.g. doctoral degree projects, to big scale projects with multiple institutions involved. Both projects with and without industry participation were brought up by the interviewees. In terms of content, the projects described ranged from research on artificial intelligence, data modelling and processing, simulation, quality and process management, robotics, circular economy, computational science, manufacturing technology and electronics.³

¹<https://github.com/pangaea-data-publisher/qualianon>

²<https://www.maxqda.com/>

³This categorization was simplified for anonymisation reasons.

2.3 Analyse the data: qualitative content analysis

In MaxQDA, the transcripts were analysed based on the qualitative content analysis method as proposed by Mayring Mayring 2022. Qualitative content analysis analyses material with regard to its content. Flick 2009 Since this material represents fixed communication (texts, images, music, etc.), the communication itself is also the subject of the analysis. It is characterised by a systematic, rule- and theory-based approach. The following section describes the practical implementation of this validation.

Modelled after Mayring Mayring 2022, a process adapted to this validation was created as shown in figure 2: The code system was formed deductively based on the theory and the question: one code each for agreement and disagreement per level and sub-step (cf. figure 3) as well as seven other codes, which are listed in table 2.

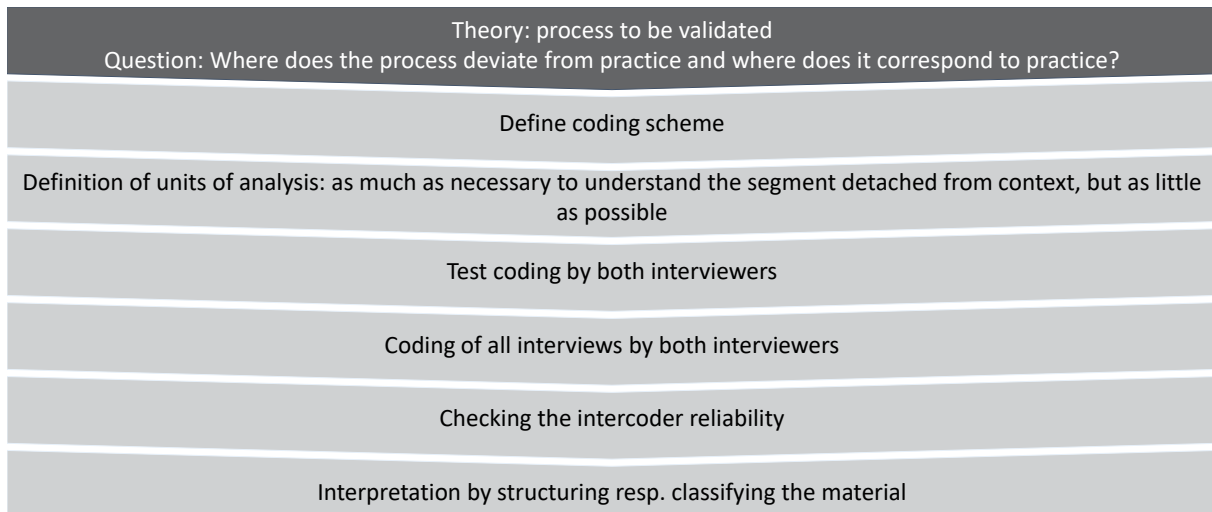


Figure 2: Content-analytical model based on Mayring 2022

A total of 66 codes were used, resulting on 741 coded segments in the ten interviews. Structuring resp. classification was used as the basic form of interpretation, i.e. filtering out certain aspects from the material by fitting its content into the code system.

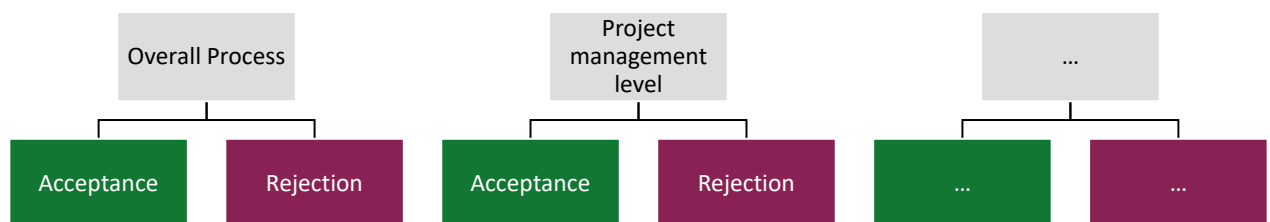


Figure 3: A sketch of the utilised code system

The coding was performed by the two interviewers individually to improve reliability and objectivity.

Afterwards, the codes were reviewed by a lead interviewer, using an intercoder analysis. In this analysis, similar coding, e.g. same codes with slightly different start and endpoints of the codes, was equalised. Additionally, conflicting codes, e.g. codes for acceptance and rejection for the same sections of transcripts, were identified. Conflicts were discussed between the reviewers and solved if possible, e.g. when a wrong coding was accidentally applied. However, only few conflicts could be solved that way. The conflicts that could not be solved in that way point towards inconsistencies in the proposed process and therefore show possible improvements. These inconsistencies and improvements are discussed in the [results and changes to the process](#).

Concluding the analysis, the coded sections were summarised by their respective codes. In that way, statements to each code could be aggregated. These aggregations are further condensed into the [results and changes to the process](#).

3 Results and changes to the process

In this section, the results are presented. Firstly, a [descriptive feedback on the interviews](#) is given. Afterwards, general feedback on the overall process is presented. Lastly, the different process levels are discussed regarding their inconsistencies in the proposed process and possible improvements. The coded segments are referred to as *statements* in the following. We have translated German statements into English.

3.1 Descriptive analysis of the interviews

The overall feedback given is positive. More than two thirds of the evaluative statements are expressing acceptance of the process in general or certain parts of it. The rest of evaluative statements express rejection. In total, about a third of all statements are indifferent, i.e. they are not giving evaluative statements on the process. These encompass the demographic descriptions of the interviewees along with their knowledge on [RDM](#) so far, descriptions of their [projects](#) used as examples, and statements on other topics on [RDM](#). Furthermore, internal annotations were made to point out passages of interest amongst the reviewers. An overview is given in Table 2.

Code Group	Acceptance	Rejection	Indifference	Total sum	group
Overall process	31	7	0	38	
Project management level	92	40	11	143	
Work package management level	74	22	8	104	
Research data management level	73	28	10	111	
Publication workflow	55	46	31	132	
Other topics (subtotal)	0	0	213	213	
Project description	0	0	46	46	
RDM -application	0	0	27	27	
Demographics	0	0	23	23	
Possible support structures	0	0	14	14	
Project structures	0	0	12	12	
Influencing factors	0	0	7	7	
Others	0	0	16	16	
Internal annotations	0	0	68	68	
Total sum	325	143	273	741	
Total sum percentages	43.9%	19.3%	36.8%	100%	

Table 2: Codes on the feedback received

3.2 General feedback on the process

The general perception of the process is very positive. 31 of 38 statements, or 81.6%, of the interviewees are positive about the fit of the process to their day-to-day work. One consensus by the interviewees was the positivity about how the structure of the process is set up from the whole [project](#) to the smallest units

of research. The division of complex problems into smaller parts is inherent to engineering sciences. One statement in particular summarised this:

“In other words, we try to break this task down into smaller subtasks. (...) This often works relatively well because this entire engineering structure is actually organised in this way. In other words, when we look at a system, I can actually always break this system down further into smaller subsystems with corresponding system boundaries etc. and actually go deeper, deeper, deeper until I just have a single element.” - Interviewee

The combination of waterfall-like structures with iterative elements was perceived as a strength of the process. Along with this, the linking of [artefacts](#) to an enriched research product were praised. Also, the completeness and end-to-end approach for research [projects](#) was acclaimed by the interviewees.

As an attention point on the proposed process, to some degree, interviewees see a discrepancy between the proposed process and the reality of research. While the process aims to depict the practice of research, it also aims to provide a *good practice* example on how the processes should look like. This however can never be expected from reality and if the process matches reality, it will rather be by chance than by careful planning, because there are many internal and external influencing factors. Some interviewees argued that these factors necessitate more iterations in the process. In contrast, others pointed out the [research proposal](#) would hold all the work packages already, which is why there would be no iteration at all on the [work package level](#). In any case, interviewees claimed that the process should not enforce a strict timeline, as this would increase overall stress.

Additionally, there were conflicting statements, when exactly research results are generated, what these are and what form they come in. In some interviewees' opinions, only research results in form of manuscripts were relevant, others saw intermediate results as results already. Furthermore, some statements hint towards results being already generated before the [research proposal](#) is accepted. These are however often not sufficient for publishing but are rather quick, but not neat solutions to prove feasibility.

Another attention point regarded the wording of certain parts of the process. For instance, while there is a [research data management level](#), still every level has its own [RDM](#) tasks. This causes unnecessary confusion and needs to be addressed. Specifically the expressions of [artefacts](#) and [findings](#) have to be concretised and enriched with examples. Similarly, parts of the proposed process should be made clearer.

Besides the overarching demarcations on the process, there were also specific statements on how certain parts of the process could be improved or do not fit the research process of the interviewees. These rejecting statements are further discussed in the following sections to improve the proposed process. Positive statements will mostly be neglected as they offer little to no impetus for improvement.

3.3 Changes to key wordings: artefact and finding

[Artefacts](#) remain the same as in our previous publication. Hamann, Robrecht, et al. 2024 However, they are specified more and implications are made explicit. Additionally, the demarcation of two [artefacts](#) are depicted more clearly.

An [artefact](#) is the result of one performance of the [artefact level](#). It is the result of a single method applied, for example: When a test bench for an experiment is designed, the plans for it are one [artefact](#) (A1). When it is build, the test bench itself is another [artefact](#) (A2), building upon the first [artefact](#) (A1). To measure something on this test bench, a software script has to be written to record data from sensors. This script is an [artefact](#) as well (A3). The recording of [raw data](#) is a new [artefact](#) (A4), resulting from [artefacts](#) (A2) and (A3). Another script (A5) has to be programmed to analyse the recorded [raw data](#) (A4). The [analysed data](#) (A6) then answers the research question. This example linking of [artefacts](#) is depicted in figure 4. How exactly [artefacts](#) are linked is depicted in figure 5.

An [artefact](#) may be a primary data collection, a physical object or the implementation of source code, or a collection of secondary data, e.g. from a systematic literature review, as it follows the definition of [research data](#) by the German Research Foundation (DFG):

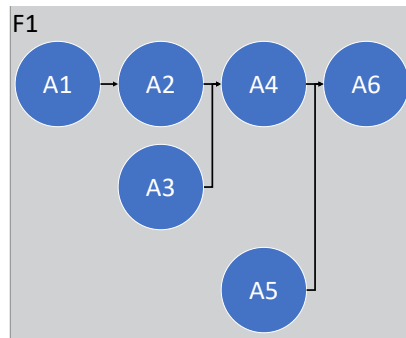


Figure 4: Linking of artefacts in the proposed process

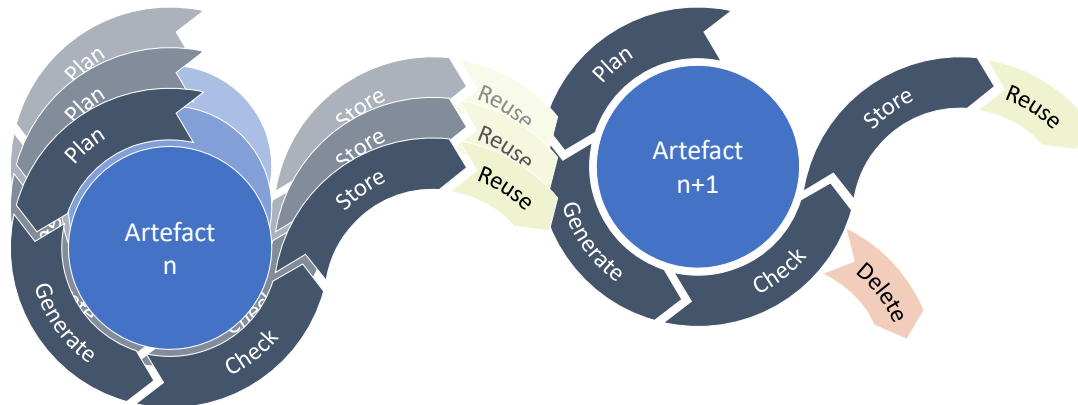


Figure 5: Procedure of **artefact** linking: Many to one relation

“ Research data includes measurement data, laboratory values, audiovisual information, texts, survey or observation data, methodological test procedures and questionnaires. Compilations, software and simulations can equally represent a central result of scientific research and are therefore also included under the term research data. Research data in some subject areas is based on the analysis of objects (such as tissue, material, rock, water and soil samples, test specimens, installations, artefacts and art objects), so its handling must be just as careful and consideration must be given to a technically adequate option for subsequent reuse whenever meaningful and possible. Should subsequent reuse of the resulting research data be closely associated with objects, then please also elaborate on this by providing all relevant information. ” German Research Foundation 2015

To elaborate further: As defined by North North 2021 with the model of the knowledge stairway, data is a number of characters (numbers, letters, special characters) that are related to each other by syntax (ordering rules). The metadata recorded along with the generation of the data adds context - i.e. a meaning - to it, so data becomes an **artefact**, which is on the information step at the knowledge stairway.

If the artefact is a physical object, the data is contained within the object, ready to be analysed and further processed in an analogue or digital way. Rixen 2026; Plomp et al. 2021; Schilthuizen et al. 2015 The biggest differences in physical data is their storage and indivisibility. An object can not be made publicly available and is stored in a physical rather than a digital storage. Yet, it has a (physical) address and access requirements that shall be clarified just like with a digital object.

No changes were made to the definition of **findings**. However, and according to North’s knowledge stairway North 2021, the **finding compilation** was more explicitly focussed on the generation of knowledge: If several pieces of information (**artefacts**) are networked and related to each other, knowledge is created. So a **finding** is knowledge, as, according to North North 2021, knowledge is a semantically interconnected information. As a finding is compiled by the interconnection of artefacts, which are bits of information, it is knowledge.

3.4 Feedback on the project management level

One of the most controversially discussed parts of the proposed process is the [project management level](#), now adjusted to [project level](#). Figure 6 shows the codes found for this level in general as well regarding certain parts of it.

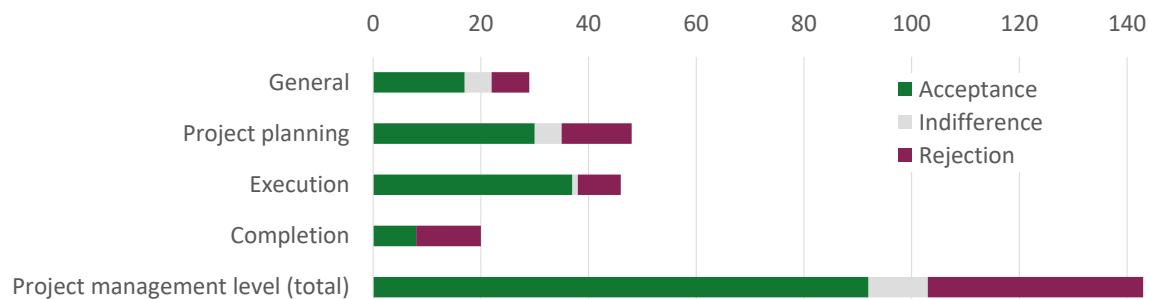


Figure 6: Codes on the feedback received on the project management level. The exact numbers can be found in table 4 in the appendix.

Most of the attention points on the [project level](#) focussed on its waterfall-like design. Some interviewees claimed, that it would still be iterative. These comments especially refer to the demarcation between [execution](#) and [completion](#). Furthermore, some interviewees annotated, that this distinction is not as clear as between [project planning](#) and [execution](#), while others emphasised it particularly as clear-cut. The acceptance of a [research proposal](#) marks the transition. A [research proposal](#) may be not just a funding proposal but any proposal made to anyone to conduct a research [project](#), e.g. to industry partners. It could also be seen as, for example, a proposal for a dissertation made to a professor. The acceptance of a [research proposal](#) or the determination of the projects completion phase can be seen as a kind of quality gate, similarly to the concept of quality gates promoted in Prefi Prefi 2014 or Hawlitzky Hawlitzky 2002: When a quality gate is reached in the process (e.g. the transition from [project planning](#) to [execution](#)), a [check](#) is performed to determine whether the activities carried out so far are sufficient (e.g. acceptance of the [research proposal](#)). If not, the quality gate cannot be passed, meaning activities must be repeated before the next phase can begin.

Every part of work performed before the acceptance is seen as [preparatory work](#): may be an early start, if the [project](#) is very likely going to be accepted, may be “quick and dirty” data, not worth of publishing, yet able to give away [findings](#). [Preparatory work](#) follows the principles of [content-wise execution](#) in its approach. Consequently, the data [generated](#) in [preparatory work](#) can be handled the same way as data created in the [execution](#). Furthermore, data used in [preparatory work](#) can also be a result of a predecessor [project](#), which means, [artefacts](#) are reused for the new [project](#)’s planning.

For the transition between [execution](#) and [project completion](#), the interviewees agreed that research is a never ending process; its completion is only achieved when the relevant demands are satisfied. There will always be something to look deeper into or reconsider or research more on. Hence, the transition to the [project completion](#) phase takes place once the supervisors for the [ongoing project management](#) determine that one or more of the following conditions are met:

- The [findings generated](#) in the [project](#) suffice to answer the overarching research question
- The [project](#)’s time frame has ended
- The [project](#) funding has run out

In the [project completion](#), the ensuring of reproducibility is an important task as it has to be seen as re-considering the work done and validating that it was done correctly: The term [ensure](#) was specified and means, that a verification and affirmation, if an activity has been performed, is conducted. If the activity was not conducted at the point of ensuring it, it has to be performed latest at this point when the ensuring takes place. Ideally, the [ensured](#) activity was performed beforehand.

In the [execution](#) phase only few attention points were brought up by the interviewees. Most importantly, the [ongoing project management](#) has to be specified. It holds not only typical tasks like comparing

the **project** plan to the current status but also encompasses the writing of intermediate reports and considerations regarding follow up **projects**. It also has to be considered that unforeseen challenges arise, demanding adaption of the original plans. The **ongoing project management** defines the next **finding** that should be **generated** based on the work done with previous **findings**, which are adapted, if needed. This decision is done at the **project** management when a **finding** is compiled.

Based on the interviewees, ensuring reproducibility can be challenging to impossible some times. Some experiments are unique and will never be able to be reproduced, e.g. the recording of a supernova⁴. Hence, the exact wording of this point has been changed. It also has to be annotated that some interviewees do not see the ensuring of reproducibility as important. This can be condensed to one statement in particular: *“What you haven’t done by then, you won’t be able to save later on either.”* Still, ensuring of reproducibility is an important task, yet coming back to the aforementioned wording of **ensure** as re-considering the work done and validating that it was done correctly.

Furthermore, the interviewees annotated that the **project completion** phase often already considers follow-up **projects**. For instance, if the prolongation of a **project** is already considered guaranteed, the **project completion** is not seen as strict as without a follow-up **project**.

Also, the interviewees pointed out that **publications** could be started at any given stage of the research **project**. Specific triggers for **publications** are discussed in section 3.7. After its redesign, the new process for the **project level** of the proposed research process is shown in figure 7.

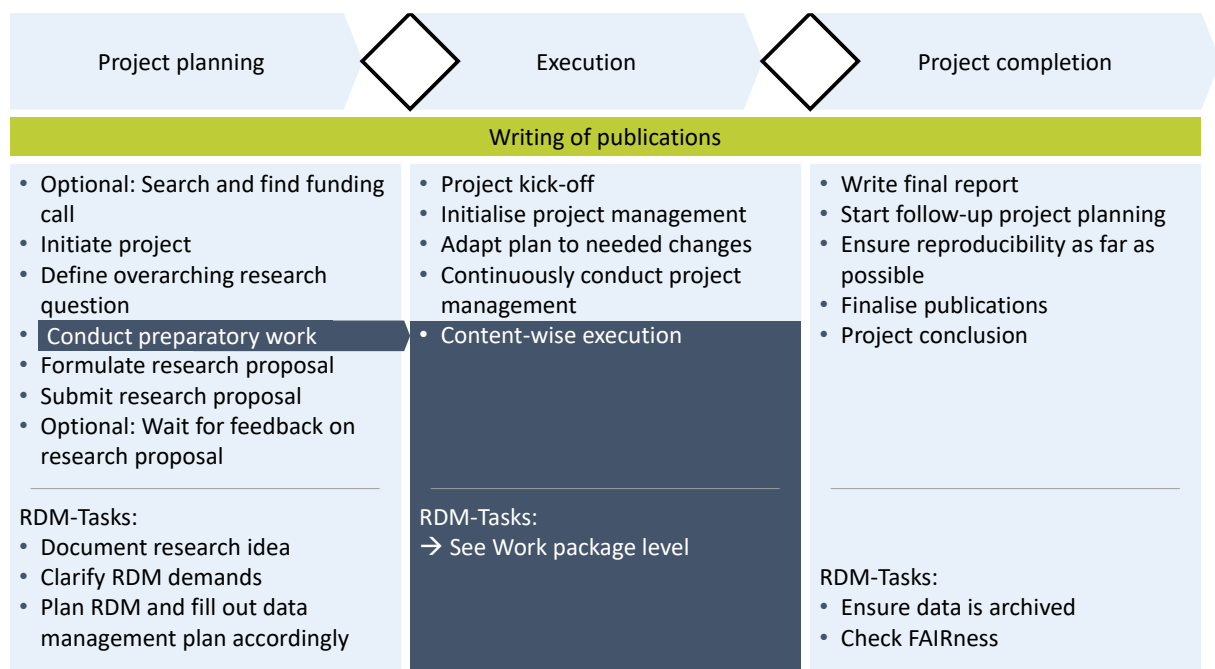


Figure 7: The redesigned project level of the proposed research process

3.5 Feedback on the work package management level

The work package management level also offers some opportunities of improvement, both in wording as well as overall process structure. The discussions with the interviewees focussed on two points in particular. On the one hand, they empathized the importance of this level being of an iterative nature, as sometimes **findings** will result in discovering new directions worth investigating. On the other hand, the **finding compilation** is seen as debatable in the proposed state. Figure 8 shows the codes found for the level.

⁴Example changed due to anonymisation.

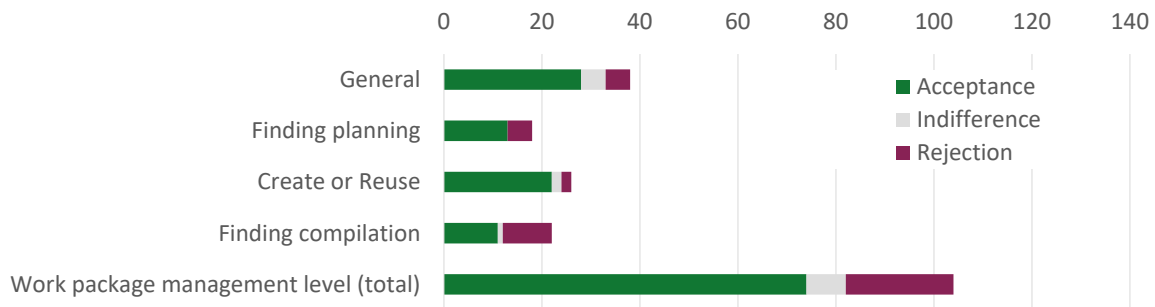


Figure 8: Codes on the feedback received on the work package management level. The exact numbers can be found in table 5 in the appendix.

The work package management level is seen under a completely new perspective: As the interviewees pointed out, the former designation does not properly represent the nature of research. Rather, the level needs to be separated from the perspective of work packages and has to focus on the [finding](#) aspect of research instead. This way, uncertainties with wording and the scope of the level are addressed: The level is now called [finding level](#).

The [finding level](#) has an iterative nature as the [findings](#) compiled at the end of the level are reported back to the [project level](#). This may either complete a work package or research question, rise a new research question, cause the need for further investigation or all of the aforementioned. It is the only level with an iterative nature. Hence, the work package perspective is now included in the [ongoing project management](#). This also specifies, that although work packages are defined in the [research proposal](#), research is still prone to unforeseen [findings](#) and changing plans during the [project](#). Furthermore, this level allows for multiple parallel [findings](#) worked on simultaneously by the same or different researchers.

The [plan approach to finding](#) step is triggered by the supervisors of the [ongoing project management](#). Firstly, a question to be answered is selected or formulated (see [plan approach to finding](#)). To answer this question, needed sub-steps are defined, each with an [artefact](#) as outcome. These [artefacts](#) are created or reused as described in section 3.3. After enough [artefacts](#) have been [generated](#) or reused, a [finding](#) can be compiled. This [finding](#) can be a significant discovery or just the fact, that the research question can not be answered with the chosen approach. In both cases, the [finding](#) is reported to the [ongoing project management](#). Optionally, a [finding](#) may be worth publishing, either as data, software or manuscript, e.g. in form of a scientific article (cf. [Feedback on publication workflow of the process](#)).

In the [finding compilation](#), [artefacts](#) are linked to each other. Following the definition of North North 2021, new knowledge is created by this linking process. As data plus its meaning (metadata) form [artefacts](#), these [artefacts](#) are information. The linking of information forms knowledge. DIN Deutsches Institut für Normung e. V. 2015 The new knowledge, whether expected, anticipated or unforeseen is reported to the [ongoing project management](#), leading to the answering of a research question or further investigation. If this knowledge is sufficient to answer the (sub-)research question selected, this (sub-)research question can be seen as solved. If the knowledge is insufficient, more [artefacts](#) are needed, and are therefore created or reused. This causes a new loop on the [finding level](#). This nature of research of having to search again for something that was not yet found was well described by one interviewee:

*“ That’s why we call it **research**. We look for something and then realise we have to look for it again because we haven’t found it. ”*

Additionally, at the beginning and the end of such a research loop on the [finding level](#), the DMP is updated to document the intention of [artefact](#) and [finding](#) creation as well as its outcome. The updated [finding level](#) is depicted in figure 9.

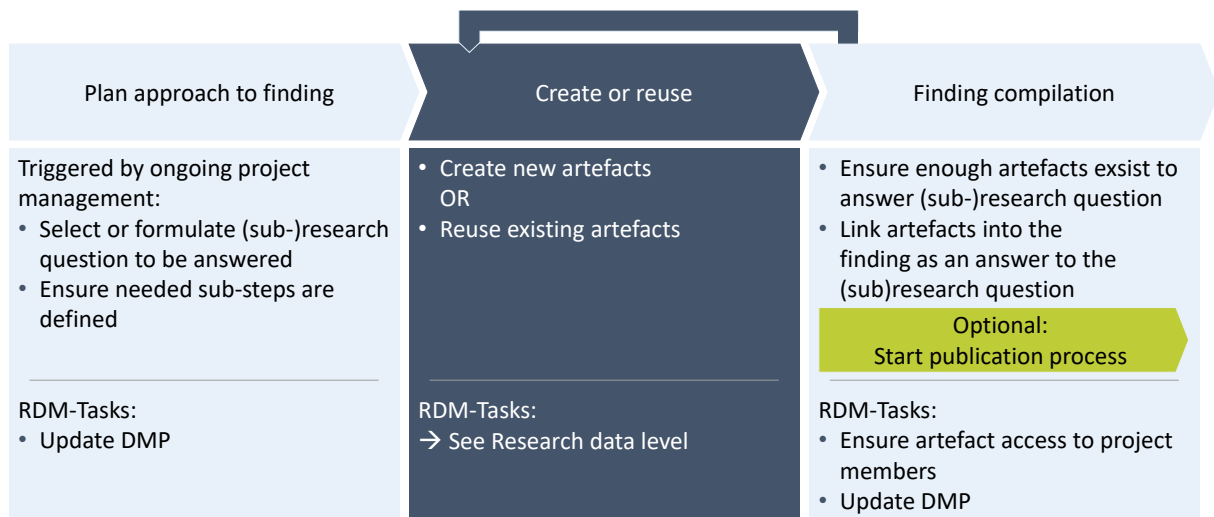


Figure 9: The redesigned finding level of the proposed research process

3.6 Feedback on the research data management level

On the [research data management level](#), interviewees mostly agreed. Their biggest concerns were clustered at the [check](#) step. This can also be seen in figure 10.

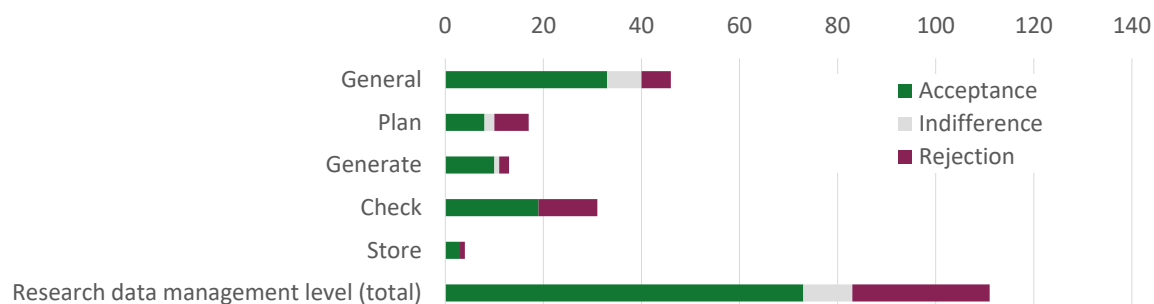


Figure 10: Codes on the feedback received on the research data management level. The exact numbers can be found in table 6 in the appendix.

The wording was adjusted to fit the [finding level](#). As the [research data management level](#) is focussed on the generation of [artefacts](#), it is renamed to [artefact level](#).

The [artefact level](#) happens to be of a very short nature in general, compared to the duration of work packages, as most activities can be completed within a few hours. Yet, the generation of data may take up to days or even weeks depending on the setup. The tasks performed by researchers are able to be performed and finished immediately and without accompanying activities. Hence, the [artefact level](#) is not iterative, but straight forward. Once performed, the outcome is either [stored](#) or deleted. In both cases the result is taken to the [finding compilation](#) to consider if more [artefacts](#) need to be [generated](#) or the new knowledge generated is sufficient to answer the research question.

In the [preparation](#) phase of the data generation, the wording is adjusted to fit the actual scope of the step, namely the [generation setup](#). The data generation is set up, meaning that the exact procedure of data generation is planned and [checked](#). This, for example might be the inspection of a test bench, if all cables are connected correctly and if the software is set up correctly for the recording of data. For interviews, this might also be the [check](#) if the recording devices are running properly, the interview guideline is present and known to the interviewers. Furthermore, input [artefacts](#) are selected. This step happens moments before the generation of data.

Then data is [generated](#). Ideally, metadata is automatically recorded with data generation. This might

encompass environmental conditions, labels for data or demographic data. The form of metadata to be used is highly dependant on the subject of investigation and the (sub-)research question to be answered.

Afterwards, the data is **checked** for correct recording, i.e. correct set up of the data generation. So this step is not a validation of data but **ensures** data correctness. Although interviewees stated that this step was often not performed, we recommend it highly to filter out unusable data right away. If the data is corrupted in any way, e.g. a sensor was not connected or malfunctioned, the data can be deleted, because no information can be drawn from this failed data generation.

If the data was recorded correctly, it can be **stored** along with the corresponding metadata. The **artefact** should be **stored** in a location that is backed up and ideally offers access to **project** members. The updated **artefact level** is shown in figure 11.

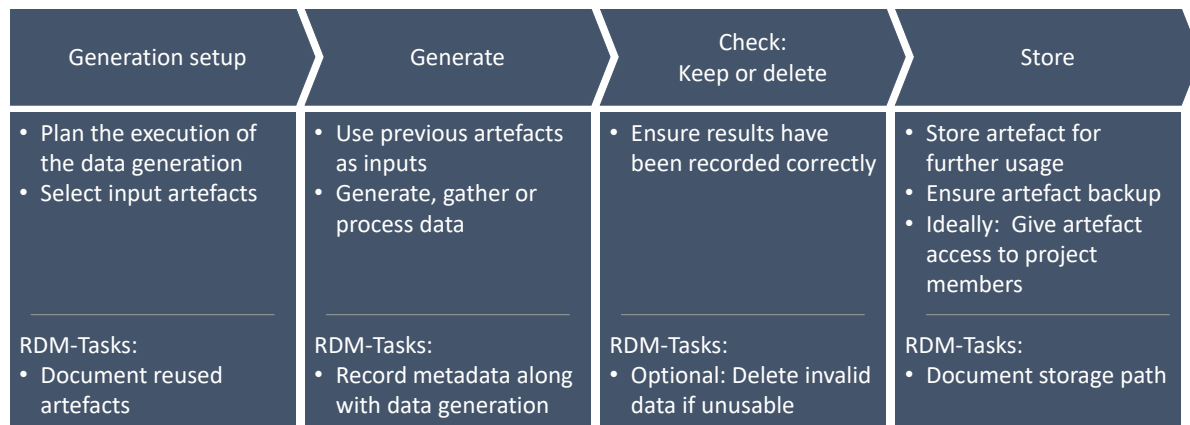


Figure 11: The redesigned research data management level of the proposed research process

3.7 Feedback on publication workflow of the process

The interviewees generally found the **publication process** to be appropriate. Nevertheless, comments such as “[the publication process] basically depicts reality as it is and should be” illustrate the discrepancy between theory and practice. For this reason, isolated differences to the model process emerged several times in the interviews. They can be attributed to time pressure and external factors (such as **project** partners or funding regulations) in day-to-day research. Figure 7 shows how this feedback was distributed over the proposed phases.

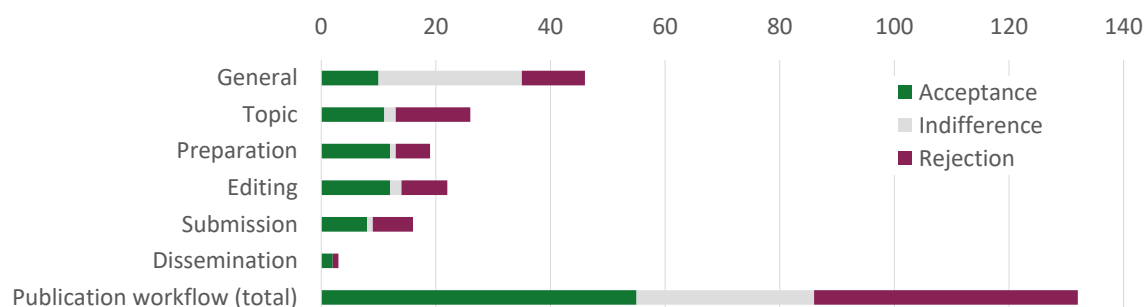


Figure 12: Codes on the feedback received on the publication workflow. The exact numbers can be found in table 7 in the appendix.

Evaluating the first two phases, a need for a clearer distinction has emerged: While the **preparation** phase sets formal framework conditions, the focus in the **topic** phase is on planning the content. So they are renamed to **topic and scoping** and **formal preparation**. In addition, a **publication** is not only triggered by the **finding compilation**, as previously stated. In practice, there are two types of triggers:

- **Self-motivated publication triggers**, e.g.: intrinsic motivation of researchers to share a **finding** that they have compiled; gaining an overview of the state of the art in the context of a review paper; turning a thesis into a paper together with the student
- **Externally motivated publication triggers**, e.g.: invitations to conferences and associated deadlines; finalising the **publication** plan from the **project** proposal; doctoral degree regulations; writing a cumulative dissertation; supervisors push for **publication**

Even if there are various triggers for **publications**, it remains clear that **finding compilation** is essential for this process:

“Let me put it this way: If I don't have any findings, then it will be difficult [...] writing a publication. But of course I can try to make findings possible in a more targeted and less targeted way. You know exactly that some things might work and then you do something in that direction, because you know that if you do this or that, then I will at least manage to generate enough new knowledge to at least justify a publication. (...) At the end of the day, of course, it would be nicer if I had a specific research question that I try to answer or was trying to answer and then say: Hey, now the research question has been answered and now I'm going to write a publication about it. Yes, there are – in practice, I would say there are both. ”

This description fits with interviewees' statements, that writing **publications** is a reflective activity, where **research data** and text production are closely linked. When converting **artefacts** and **findings** into a written form (to create a medium that passes on the knowledge from the **finding compilation** to other people), gaps may become visible that need to be filled with additional content, or new contexts may emerge.

“I always find that it all makes a lot of sense in my head. But it's only when you put it down on paper that you realise: Oh no, that's somehow, there's no thread running through it. So that's actually always a point where you have to think more deeply about it when you're writing what have you actually just done here, because you have to get your own thoughts down somehow, and so there's a lot of reflection involved. ”

At the same time, the interviewees emphasised that there are also **publication** types such as concept papers that contain less data. Interviewees stated that **publications** could occur at any time over the course of the whole **research project**. Another aspect that the interviewees repeatedly mentioned was preprints: By publishing **publications** that have not yet been peer-reviewed, they want to position themselves quickly in the dynamic academic discourse and make their work referencable so that they can be followed up in subsequent **publications**.

Finally, the interviewees addressed the communication of scientific results to the public - usually referred to as “**science communication**” - which is becoming increasingly established in the job description of researchers Könniker 2012. Communication within the scientific system is correspondingly “**scholarly communication**”. (cf. e.g. Hagenhoff et al. 2007, Burns et al. 2003) Since these two sub-areas of publishing differ in terms of target group and communication goal and form, but are similar looking at the basic process, the revised graphic is divided into two parts, as shown in figure 13.

The content on **science communication** is essentially based on the explanations of Könniker Könniker 2012, Frick and Seltmann Frick and Seltmann 2024 and Bertemes et al. Bertemes et al. 2024. A key difference between **science** and **scholarly communication** lies in the support. Depending on the involvement of the respective institution's communication department, process steps in the area of **science communication** (e.g. parts of **editing** or **submission**) are handled by the department rather than by the researchers themselves - in contrast to **scholarly communication**.

In addition to the highlighted aspects so far, the **publication process** has undergone further minor adjustments, such as the addition of internal reviews and (external) approval processes.

3.8 Overall process

The overall process was met with general approval by the interviewees. Still, the refinements made to the process on the basis of the interviewees feedback improved it significantly. Wordings have been specified, steps and tasks have been made more explicit and the structure of the process was adapted to



Figure 13: The redesigned publication process

the reality of research while also upholding standards of RDM. The final process is depicted in figure 14.

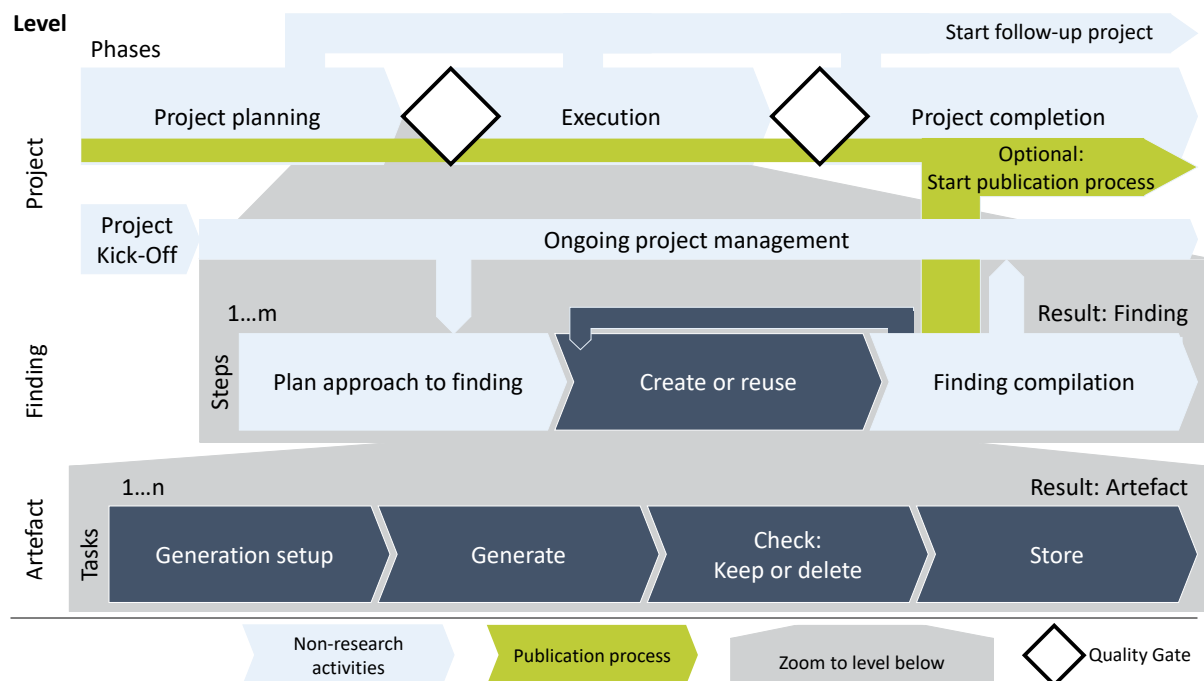


Figure 14: General structure of the proposed RDM process

In comparison to figure 1, the biggest changes are:

1. The option to start follow-up projects in any of the process' phases
2. The option to start a publication process in any of the process' phases
3. The option to generate findings in the project planning phase
4. The project kick-off was moved to not be included in the content-wise execution and instead precede the ongoing project management.
5. A feedback loop between finding compilation and plan approach to finding
6. A feedback loop between finding compilation and create or reuse
7. Wording changes, especially on the names of the levels:
 - Project management level changed to project level
 - Work package level changed to finding level
 - Research data management level changed to artefact level

This process validated the previously identified needs, demands, and practices of researchers from engineering sciences. It aims to resemble the day-to-day research activities of researchers in engineering sciences as best as possible while also integrating RDM activities as seamlessly as possible.

4 Discussion and Limitations

The answer to the research question can be given with a high certainty. The evidence for this statement is high, as it is based on the consolidation of eight problem-centred interviews with 19 researchers, the surveying of 13 research projects by another eight researchers on their workflows and this validation with another ten researchers in expert interviews. This total of 37 different researchers questioned on their research's demands and structure.

While the first two methods were biased on the RWTH and Leibniz University Hannover (LUH), the validation took into consideration the whole TU9. While, smaller universities or universities of applied sciences might have different approaches to research, the approach of validation via TU9 allows for insights into a big part of engineering sciences.

Furthermore, the changes and adaptations to the process, while seemingly big, are not crucial. The biggest changes were made due to wording inconsistencies. Certain parts of the process were explicated and defined more precisely, or in some cases, more openly to allow for a better situational fit. The overall process structure with its levels and their contents remain unchanged.

The validation, which was urgently missing in the last paper, was now performed. It confirmed the good reception the process got in several events where it was presented (see Hamann and Moser 2022; Hamann 2023a; Hamann 2023b; Hamann and Werheid 2023). Overall, it affirmed the process with minor changes made to it.

Nevertheless, the methods used should be critically contextualized: The usual quality criteria such as objectivity, reliability and validity in the sense of definitions from natural sciences can only be applied to expert interviews to a limited extent. For example, the interviews themselves take place in social situations that create a dynamic between interviewers and interviewees or even unintentional mutual influences, which can lead to bias. Bogner et al. 2014 The objectivity of qualitative content analyses - a method which core is the interpretation of the material by people with different knowledge and views - can at least be strengthened by instruments such as intercoder analysis. Mayring 2022

5 Conclusion and Outlook

In this article, we validated our proposed process for **project-oriented RDM** in engineering sciences. This process, now called Project- and RDM-Oriented Research Process for Engineering Sciences (PRO Research Process 4Ing), is the result of a thorough investigation of the problems, requirements, workflows and practices of engineering researchers. In total 37 different researchers were consulted by the means of different methods.

PRO Research Process 4Ing is validated in this article by a series of ten expert interviews. The result is a validated process that received minor adjustments to better fit the reality of day-to-day work and facilitate understandability. This also leads to the answer to the proposed research question:

***Can our proposed **RDM** process be deployed into engineering research projects?
With the changes made to it, yes.***

PRO Research Process 4Ing will provide the foundation of Jarves, the Joint Assistant for Research in Versatile Engineering Sciences ⁵. This digital data steward guides engineering researchers in their **RDM** throughout their whole research **project**. Jarves will also contain the management of **RDM** guidelines within a decision support system. Hamann, Robrecht, et al. 2024 Jarves was validated by a Proof-of-Concept-Study in a real world application by Hamann et al. Hamann, Florides, et al. 2025.

PRO Research Process 4Ing offers a missing link between theoretical **RDM** and actual research practices. It offers a bridge between research **projects** and their structure and the life cycles of **research data**. We hope, that PRO Research Process 4Ing facilitates the application of **RDM** in engineering sciences. Additionally, in future research PRO Research Process 4Ing could be transferred to other disciplines that utilize **project-oriented** research.

A Appendix

This appendix contains a glossary on the most important terms in this paper and the process it describes as well as further elaborations on the exact numbers of codes for each level of the process.

⁵<https://jarves.nfdi4ing.de/>

A.1 Glossary

This glossary aims to facilitate understandability for those who would like to implement, utilise or further develop this process. The glossary is an updated version of the previous paper's glossary. Hamann, Robrecht, et al. 2024

Analysed data	Any data that is processed in any way that allows for new information to be derived from it. On artefact level. Straight forward.
Artefact	The result of one iteration of the artefact level . An artefact is planned, contextualised and validated research data in its smallest unit. It may be a primary data collection or the implementation of source code. It was generated by creating a generation setup , generating the data and performing a check to ensure the correct recording and, if valid, storing the data. An artefact is information. On artefact level. Straight forward.
Artefact level	The level on which artefacts are generated. It contains the tasks generation setup , generate , check and store . It has a straight forward structure with distinct tasks following one after another. On artefact level. Straight forward.
Check	Check the generated data for corruption. This task ensures data correctness. On artefact level. Straight forward.
Compile finding	Term from the previously designed process. See now finding compilation . On finding level. Iterative.
Completion	Term from the previously designed process. See now project completion . On project level. Waterfall structure.
Content-wise execution	All activities on the finding level containing the generation of findings and knowledge. On finding level. Iterative.
Core research activities	All activities on the artefact level containing the generation of research core data . On artefact level. Straight forward.
Create or reuse	The step of creating a new artefact or reusing an existing one by importing it into the current project . The creation of a new artefact can be done by using any amount of existing artefacts. On artefact level. Straight forward.
Dissemination	Last of the five publication process phases. Dissemination includes all activities that promote the distribution of a publication , such as inclusion in the personal, publicly accessible publication list, social media posts or even personal recommendations via mailing lists. On publication process. Straight forward.
Editing	The third of the five publication process phases. Can only be started if the topic and scoping phase is finished. Includes all activities that revolve around the actual production of text. On publication process. Iterative.
Ensure	Ensuring an activity has been performed and, if not, performing it latest at this point when the ensuring takes place. Ideally, what is ensured was performed beforehand.
Execution	The second of the three research project phases. Can only be started if the acceptance of the research proposal . On project level. Waterfall structure.
Externally motivated publication trigger	Trigger for publication initiated by the external parties or factors, e.g.: invitations to conferences and associated deadlines, finalising the publication plan from the project proposal, doctoral degree regulations, writing a cumulative dissertation, supervisors push for publication

Finding	The result of one iteration of the finding level . It may be the result of one work package or the answer to one research question. It was generated by planning what should be found out, compiling one or several artefacts and hereby deriving new knowledge from this process. A finding is knowledge. On finding level. Iterative.
Finding compilation	Compile a finding from artefacts . In this way, new knowledge is generated. On finding level. Iterative.
Finding level	The level on which findings are generated. It contains the steps plan approach to finding , create or reuse and finding compilation . On finding level. Iterative.
Finding planning	Term from the previously designed process. See now plan approach to finding . On finding level. Iterative.
Formal preparation	Second of the five publication process phases. Formal preparation includes all activities that relate to the formal framework conditions of the publication process and trigger corresponding planning. On publication process. Straight forward.
Generate	In the generate task, new artefacts are brought into existence. This can happen by using existing artefacts or without them. New data is generated, gathered or gained by processing of existing artefacts . Due to the annotation of the data with metadata, meaning is provided, adding meaning to the data and therefore turning it into information. On artefact level. Straight forward.
Generation setup	The data generation setup happens immediately before the generate task. It is the planning and affirmation of the exact procedure of data generation. Furthermore, input artefacts are selected. On artefact level. Straight forward.
Ongoing project management	Over the course of the project , the ongoing project management tracks the process of the project in accordance to the project plan. The ongoing project management defines the next finding that should be generated based on the work done with previous findings, which are adapted, if needed. It also contains other project management tasks, such as the writing of reports and project meetings. On project level. Waterfall structure.
Plan	Term from the previously designed process. See now generation setup . On artefact level. Straight forward.
Plan approach to finding	When the supervisors trigger the investigation of a specific topic, a corresponding research question is either selected (if existent, e.g. in the proposal) or defined (if not existent yet). Additionally, needed sub-steps to answer the research question are defined. On finding level. Iterative.
Planning of the data generation	Term from the previously designed process. See now generation setup . On artefact level. Straight forward.
Preparation	Term from the previously designed process. See now formal preparation . On publication process. Straight forward.
Preparatory work	Preparatory work is content-wise work conducted in a research project before the research proposal is submitted. In preparatory work, content-wise execution can already be conducted and hence, findings can already be generated. On project level. Waterfall structure.
Project	Unique activity with a defined outcome and predetermined resources in terms of costs and time Möller and Dörrenberg 2003. On project level. Waterfall structure.

Project completion	The last of the three research project phases. Can only be started if the execution is finished. On project level. Waterfall structure.
Project kick-off	At the beginning of the execution phase, this step occurs to actually start the project after the sometimes long time passed since the project planning phase. Here, the project team is formed if not yet existent. Also, the proposal is consulted to get an overview over the goals and needed approach. On finding level. Straight forward.
Project level	The level on which projects are conducted. It contains the phases project planning , execution and project completion . It has a waterfall-like structure with distinct phases. On project level. Waterfall structure.
Project management level	Term from the previously designed process. See now project level . On project level. Waterfall structure.
Project-oriented research	Research with a planned and defined outcome and predetermined resources in terms of costs and time (c.f. Möller and Dörrenberg 2003).
Project planning	The first of the three research project phases. The research project starts with this phase. On project level. Waterfall structure.
Publication	A publication is a written record of a finding . The knowledge that is generated during the finding compilation can be passed on to other people through this medium.
Publication process	Independently running process that can start at any given stage of the research project . Its result is a publication that contains the knowledge generated during the finding compilation .
Raw data	Any data recorded or created, but not yet analysed. Does not generate any new knowledge. On artefact level. Straight forward.
Research core data	The research data collected in a research project , which generates or is used to generate new knowledge. On artefact level. Straight forward.
Research data	“Research data includes measurement data, laboratory values, audiovisual information, texts, survey data, objects from collections or samples that are created, developed or analysed in the course of scientific work. Methodological test procedures such as questionnaires, software and simulations can also represent key results of scientific research and should therefore also be categorised as research data” German Research Foundation 2015. On artefact level. Straight forward.
Research data management (RDM)	The handling of research data (e.g. collection, organization, storage, and documentation) during and after a research process.
Research data management level	Term from the previously designed process. See now artefact level . On artefact level. Straight forward.
Research project	Project to investigate a specific research question. On project level. Waterfall structure.
Research project data	The data used to describe the information about a research project without the creation of new knowledge.
Research proposal	A proposal for a research project . This can but does not need to be a funding proposal. It can also be a proposal of a dissertation project to a professor or a proposal of an industry project to an industry partner. On project level. Waterfall structure.
Scholarly communication	Scholarly communication is a type of communication in the publication process . It takes place within the scientific system by researchers who disseminate their findings peer to peer. On publication process.

Science communication	science communication is a type of communication in the publication process . Researchers communicate their findings to the public, possibly with the support of communications departments. On publication process.
Self-motivated publication trigger	Trigger for publication initiated by the researcher, e.g.: intrinsic motivation of researchers to share a finding that they have compiled, gaining an overview of the state of the art in the context of a review paper or turning a thesis into a paper together with the student.
Store	When the artefact was checked , it can be stored. Ideally, it is stored in a way that project members can access it. On artefact level. Straight forward.
Submission	This phase comprises the activities surrounding the submission of the publication , e.g. to the publisher. The core is the peer review, which ensures scientific quality. On publication process. Iterative.
Topic	Term from the previously designed process. See now topic and scoping . On publication process. Straight forward.
Topic and scoping	This phase is triggered by either a self-motivated publication trigger or a externally motivated publication trigger . The availability of a finding is regardless of this crucial. In this phase, the content of the publication is determined. On publication process. Straight forward.
Work package level	Term from the previously designed process. See now finding level . On finding level. Iterative.

A.2 Previously designed process

This section of the appendix describes the original process proposed in Hamann, Robrecht, et al. [2024](#) before its validation. It combines three levels of research, along with waterfall [project](#) management and agile approaches. Besides the relevant literature, it is also based on eight problem-centred interviews to gather requirements of researchers regarding the application of [RDM](#). Furthermore, eight engineering researchers were interviewed on their research and typical workflows. The result is the three layered, multi-staged process as shown in figure [15](#).

The [project management level](#) encompasses a waterfall-like approach. The [project planning](#) includes all activities that happen before the research [project](#) is accepted, either in form of a provided grant or accepted [research proposal](#). Then the [project](#) is executed. This [execution](#) includes the [content-wise execution](#), [ongoing project management](#) and other activities. The [project completion](#) starts when the [project](#) is coming to an end. It contains activities such as the writing of final reports or archiving of data. This level of the process is depicted in greater detail in figure [16](#).

Below the [project management level](#), the [work package level](#) is placed, where work packages are processed. Hence, this level is conducted n times in an iterative approach. For [finding planning](#), firstly, a work package is chosen and divided into sub-steps needed to reach its goal. Afterwards, [artefacts](#) are created or reused - what researchers consider "actual research" and happens on the [research data management level](#). The creation of [artefacts](#) has to happen n times to finish the work package. Afterwards, the so created [artefacts](#) are compiled to a [finding](#) in the [finding compilation](#) step. The [finding](#) is the result of the work package. The work package management level is depicted in figure [17](#).

[Artefacts](#) are the smallest unit of data in a research [project](#). Depending on the use case, it can be a single file, a data set of several files or physical object, e.g. a demonstrator or a workpiece. If it is a digital piece of data, it is considered to be a digital object. Schwarzmänn [2020](#) However, it is not necessarily a FAIR Digital Object (FAIR-DO) by the definition of Schwarzmänn Schwarzmänn [2020](#), as a Persistent Identifier (PID) might be missing. An artefact can also be a real, physical object, as objects themselves can contain information that is stored within the object and can be analysed. Rixen [2026](#); Plomp et al. [2021](#); Schilthuizen et al. [2015](#) An [artefact](#) is always the single outcome of one research method applied.

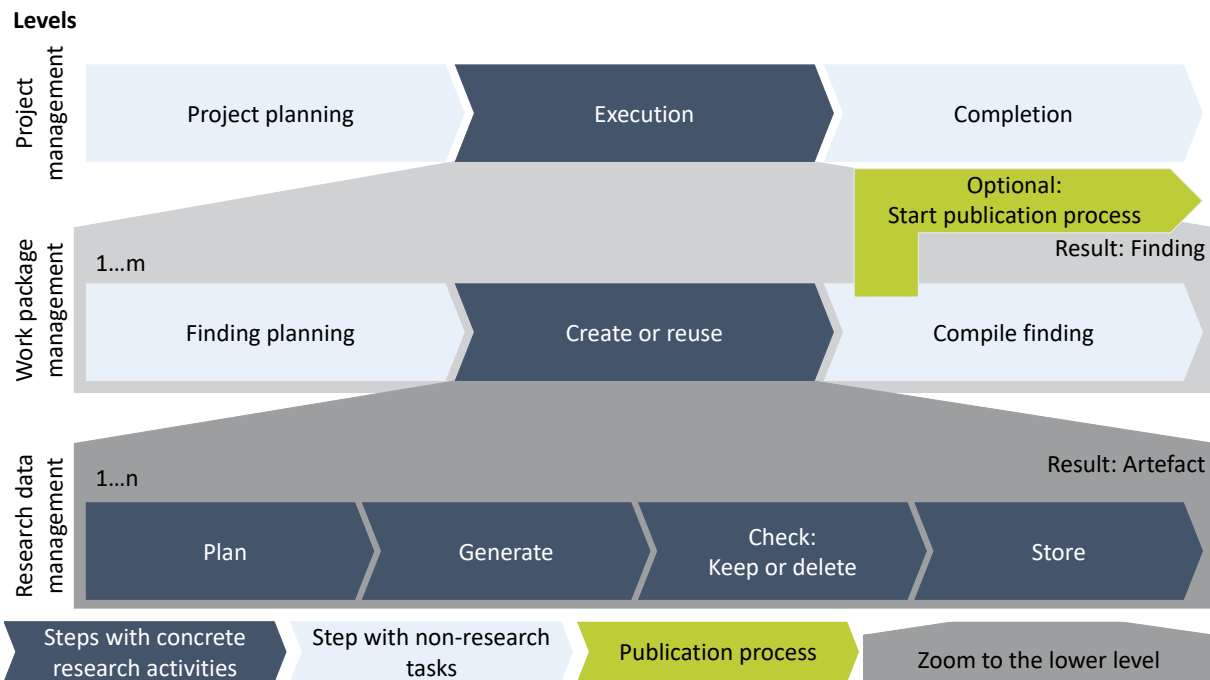


Figure 15: General structure of the previously designed RDM process Hamann, Robrecht, et al. 2024

On the **research data management level**, each **artefact** is created by firstly planning its generation. This planning happens immediately before it is **generated**. It is not an experimental design, rather the experimental design would be a previous **artefact**. Instead, the **planning of the data generation** encompasses the gathering of the needed **artefact**, the methodical setup and the **check** if everything is properly configured. Then, the data is **generated**. Optimally, metadata is recorded along with the data generation, annotating the data, transferring it from data to data with context. Afterwards, a **check** of the data **ensures** the data was recorded at all, is reasonable and within expectations and not corrupted. Otherwise, it can be deleted immediately. If the data was recorded properly, it can be **stored** for further usage. This process revolves around each **artefact**, as shown in figure 5 and in figure 18.

Additionally to the **finding compilation**, a **publication process** can be started. It starts optionally from the **work package level** and runs in parallel. The **topic** definition, the first step in the process, is determined by the **finding compilation**. The **publication** is then prepared, which includes, for example, finding a suitable medium. This is followed by **editing**: **Artefacts**, **findings** and the knowledge generated as a result are converted into text and - as an RDM task - published. In the **submission** step, the **publication** is submitted, subjected to a quality check through a peer review process and finally published. This is at last followed by **dissemination** as illustrated in figure 19.

The goal of the developed process is the combination of **project-oriented** research, demands of engineering researchers and **RDM** processes. This combination could solve the problem of lacking applicability of **RDM** found in Hamann, Robrecht, et al. 2024. However, it has to be validated, if this goal could be reached by the proposed process.

Parts of the proposed process have also already been taken up and further investigated and implemented by Wawer et al. Wawer et al. 2024. However, in their paper, Wawer et al. Wawer et al. 2024 focus on the **research data management level**, elaborating on the concept of **artefacts**, their linking and compilation into a research result. In addition to their process, Wawer et al. Wawer et al. 2024 point out, "the context of RDM should be established at **project level**, as data is usually archived at the end of the project and a data management plan is created at the start of the project". This establishment is the focus of our proposed process both in this and the previous paper. Yet, to this point the process has not been proven to be sufficient, which makes a validation of this process necessary as presented in the next section.

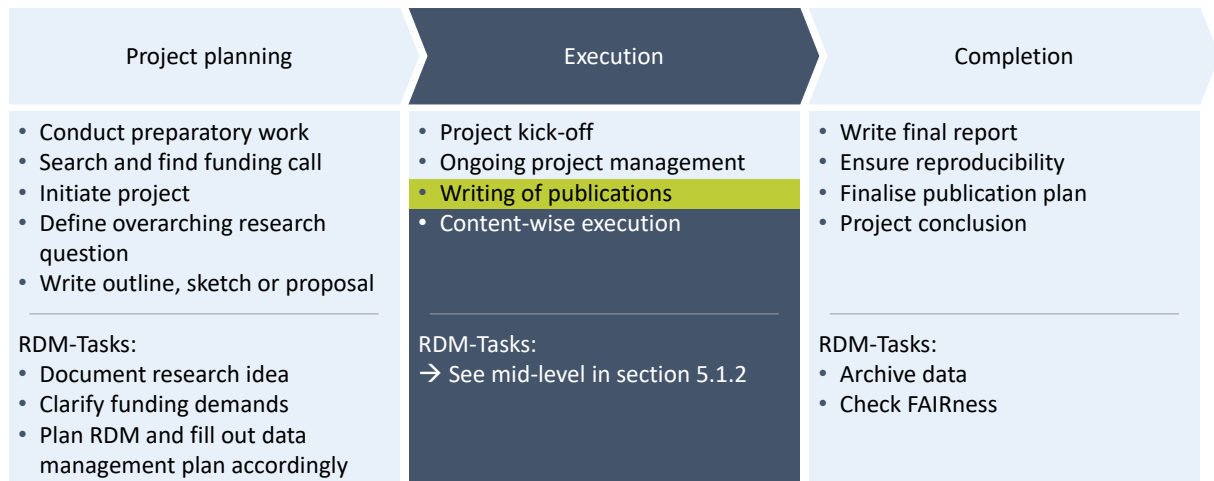


Figure 16: Project management level of the proposed research process Hamann, Robrecht, et al. 2024

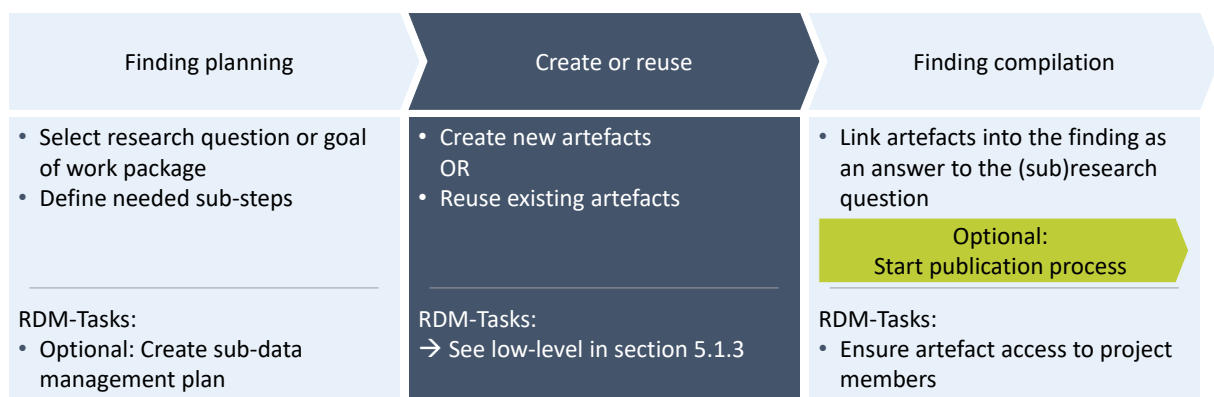


Figure 17: Work package management level of the proposed research process Hamann, Robrecht, et al. 2024

A.3 Tables on Codes

This section of the appendix further elaborates on the exact numbers of codes gathered for each of the processes parts. Table 4 shows codes gathered for the project management level.

Table 5 depicts the number of codes on the feedback received on the work package management level.

Table 6 depicts the number of codes on the feedback received on the research data management level.

Table 7 depicts the number of codes on the feedback received on the publication workflow.

Plan	Generate	Check: Keep or delete	Store
- Plan the execution of the data generation - Select input artefacts	- Use previous artefacts as inputs - Generate, gather or process data	Check results for validity	Save artefact for further usage
RDM-Tasks: Document reused artefacts	RDM-Tasks: Record metadata along with data generation	RDM-Tasks: Optional: Delete invalid data if unusable	RDM-Tasks: Document storage path

Figure 18: Research data management level of the proposed research process Hamann, Robrecht, et al. 2024

Topic	Finding compilation triggers a publication process
Preparation	- Includes steps such as: open access and open data strategy, publication medium, export control, literature review in addition to the previous work - Synergies with other steps in the research process, especially formal requirements such as funding regulations
Editing	- Convert artifacts, findings and the knowledge generated into text and text-supplementing elements - Derive the methodology to be described from the previous steps of data generation - Publish and reference artifacts and findings
Submission	- Includes submission to the publication medium, peer review and subsequent revision, final publication - Content is more comprehensible for peers because the research data is available and its provenance is clear.
Dissemination	- Disseminate publication in accordance with funding requirements - Particularly target group-oriented communication by sharing text and/or research data

Figure 19: The publication process Hamann, Robrecht, et al. 2024

Code Group	Acceptance	Rejection	Indifference	Total sum	group
Project management level (total)	92	40	11	143	
Comprised of:					
General	17	7	5	29	
Project planning	30	13	5	48	
Execution	37	8	1	46	
Completion	8	12	0	20	

Table 4: Codes on the feedback received on the project management level

Code Group	Acceptance	Rejection	Indifference	Total sum	group
Work package management level (total)	74	22	8	104	
Comprised of:					
General	28	5	5	38	
Finding planning	13	5	0	18	
Create or Reuse	22	2	2	26	
Finding compilation	11	10	1	22	

Table 5: Codes on the feedback received on the work package management level

Code Group	Acceptance	Rejection	Indifference	Total sum	group
Research data management level (total)	73	28	10	111	
Comprised of:					
General	33	6	7	46	
Plan	8	7	2	17	
Generate	10	2	1	13	
Check	19	12	0	31	
Store	3	1	0	4	

Table 6: Codes on the feedback received on the research data management level

Code Group	Acceptance	Rejection	Indifference	Total sum	group
Publication workflow (total)	55	46	31	132	
Comprised of:					
General	10	11	25	46	
Topic	11	13	2	26	
Preparation	12	6	1	19	
Editing	12	8	2	22	
Submission	8	7	1	16	
Dissemination	2	1	0	3	

Table 7: Codes on the feedback received on the publication workflow

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